SAMPLE

String Theory

Vibrating Strings and Hidden Dimensions

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Introduction

Replace the familiar picture of particles as dimensionless points with tiny one-dimensional filaments, and you have the central move of this book's subject: a framework in which every elementary particle is a particular note played by a vibrating string. The replacement sounds modest, but it changes everything downstream, because among the allowed notes there is inevitably one whose properties match the graviton, the hypothetical messenger of gravity. Any framework with that feature is, at minimum, a candidate quantum theory of gravity, and reconciling gravity with quantum mechanics is the oldest unfinished business of theoretical physics.

The subject's history is layered. It began in the late 1960s as an attempt to describe the strong nuclear force, failed at that job for instructive reasons, and was rescued in 1974 when a few researchers realized the 'failure' was actually a gravity-shaped success in disguise. Through the 1980s five separate consistent formulations crystallized; in the mid-1990s Edward Witten conjectured that the five were different limits of a single deeper structure in eleven dimensions, called M-theory, and the field's center of gravity shifted to questions about what that deeper structure is.

The dividends have not been only gravitational. Techniques invented for strings have been borrowed across physics and mathematics: they have illuminated black hole entropy, sharpened tools in nuclear physics and condensed matter physics, and produced genuinely new mathematics, some of it unexpected enough that physicists discovered results mathematicians had not yet imagined. A striking instance is the dictionary, uncovered in the late 1990s, between a gravitational world and a particle world living on its boundary -- a correspondence that has become a working instrument in half a dozen fields.

None of this has silenced the critics, and the book that follows will not silence them either, only present the case honestly. The framework has no confirmed experimental prediction to its name; its mathematics is ferocious; and its internal logic leads to a staggering multitude of possible universes, which raises uncomfortable questions about what kind of theory it is and how it could ever be tested. Whether those difficulties are growing pains or symptoms is the live question of the subject, and it is the question the chapters ahead take up in order: how the idea arose, what it actually says, what has been built from it, and where the argument over its future now stands.

Chapter 1: The Discovery of String Theory

Five consistent superstring formulations took shape before the mid-1990s conjecture that they were all different limits of one eleven-dimensional parent theory, M-theory. In late 1997 came a second landmark: a dictionary, now called the AdS/CFT correspondence, relating a gravitational string world to an ordinary quantum field theory living on its boundary. A recognized shortcoming persists alongside these successes, namely that the complete non-perturbative definition of the theory is still missing.

Another issue is that the theory is thought to describe an enormous landscape of possible universes, which has complicated efforts to derive realistic particle-physics models from the framework. Detractors inside the community have taken both facts as grounds for skepticism about the whole unification program. In the 20th century, two theoretical frameworks emerged for formulating the laws of physics.

The first is Albert Einstein's general theory of relativity, a theory that explains the force of gravity and the structure of spacetime at the macro-level. The other is quantum mechanics, a completely different formulation, which uses known probability principles to describe physical phenomena at the micro-level. By the late 1970s, these two frameworks had proven to be sufficient to explain most of the observed features of the universe, from elementary particles to atoms to the evolution of stars and the universe as a whole.

In spite of these successes, there are still many problems that remain to be solved. One of the deepest problems in modern physics is the problem of quantum gravity. The general theory of relativity is formulated within the framework of classical physics, whereas the other fundamental forces are described within the framework of quantum mechanics.

A quantum theory of gravity is needed in order to reconcile general relativity with the principles of quantum mechanics, but difficulties arise when one attempts to apply the usual prescriptions of quantum theory to the force of gravity. String theory is a theoretical framework that attempts to address these questions. The starting point for string theory is the idea that the point-like particles of particle physics can also be modeled as one-dimensional objects called strings.

String theory describes how strings propagate through space and interact with each other. In a given version of

string theory, there is only one kind of string, which may look like a small loop or segment of ordinary string, and it can vibrate in different ways. On distance scales larger than the string scale, a string will look just like an ordinary particle consistent with non-string models of elementary particles, with its mass, charge, and other properties determined by the vibrational state of the string.

String theory's application as a form of quantum gravity proposes a vibrational state responsible for the graviton, a yet unproven quantum particle that is theorized to carry gravitational force. One of the main developments of the past several decades in string theory was the discovery of certain 'dualities'--mathematical transformations that identify one physical theory with another.

Physicists studying string theory have discovered a number of these dualities between different versions of string theory, and this has led to the conjecture that all consistent versions of string theory are subsumed in a single framework known as M-theory. Studies of string theory have also yielded a number of results on the nature of black holes and the gravitational interaction.

There are certain paradoxes that arise when one attempts to understand the quantum aspects of black holes, and work on string theory has attempted to clarify these issues. In late 1997 this line of work culminated in the discovery of the anti-de Sitter/conformal field theory correspondence (AdS/CFT for short). This is a theoretical result that relates string theory to other physical theories which are better understood theoretically.

The AdS/CFT correspondence has implications for the study of black holes and quantum gravity, and it has been applied to other subjects, including nuclear and condensed matter physics. Since string theory incorporates all of the fundamental interactions, including gravity, many physicists hope that it will eventually be developed to the point where it fully describes our universe, making it a theory of everything.

One of the goals of current research in string theory is to find a solution of the theory that reproduces the observed spectrum of elementary particles, with a small cosmological constant, containing dark matter and a plausible mechanism for cosmic inflation. While there has been progress toward these goals, it is not known to what extent string theory describes the real world or how much freedom the theory allows in the choice of details.

One of the challenges of string theory is that the full theory does not have a satisfactory definition in all circumstances. The scattering of strings is most straightforwardly defined using the techniques of perturbation theory, but it is not known in general how to define string theory nonperturbatively. It is also not clear whether there is any principle by which string theory selects its vacuum state, the physical state that determines the properties of our universe.